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Why Would Anybody Think This?

How Quantum Mechanics Came to Be

"Sometimes I've believed as many as six impossible things before breakfast," notes the White Queen to Alice in *Through the Looking Glass*. That can seem like a useful skill as one comes to grips with quantum mechanics in general, and Many-Worlds in particular. Fortunately, the impossible-seeming things we're asked to believe aren't whimsical inventions or logic-busting Zen koans; they are features of the world that we are nudged toward accepting because actual experiments have dragged us, kicking and screaming, in that direction. We don't choose quantum mechanics; we only choose to face up to it.

Physics aspires to figure out what kinds of stuff the world is made of, how that stuff naturally changes over time, and how various bits of stuff interact with one another. In my own environment, I can immediately see many different kinds of stuff: papers and books and a desk and a computer and a cup of coffee and a wastebasket and two cats (one of whom is extremely interested in what's inside the wastebasket), not to mention less solid things like air and light and sound.

By the end of the nineteenth century, scientists had managed to distill every single one of these things down to two fundamental kinds of substances: *particles* and *fields*. Particles are point-like objects at a definite location in space, while fields (like the gravitational field) are spread throughout space, taking on a particular value at every point. When a field is oscillating across space and time, we call that a "wave." So people will